Towards Object-Centric Process Mining for Personal Health Management

Maria Inês Ribeiro[1][0009-0001-7746-4685], Laura Genga[1][0000-0001-8746-8826], Monique Simons[2][0000-0002-6475-4616] and Pieter Van Gorp[1][0000-0001-5197-3986]

[1] Eindhoven University of Technology, Het Eeuwsel 53, 5612AZ Eindhoven, Netherlands
[2] Wageningen University & Research, Hollandseweg 1, 6706KN Wageningen, Netherlands
m.i.da.graca.jorge.da.silva.ribeiro@tue.nl

Abstract. Wearable and mobile technologies are transforming personal health management by enabling continuous monitoring and tailored interventions for behavior change. As digital health advances toward personalization, understanding how personal health behaviors evolve and interact in dynamic contexts is critical. Current data-driven methods rely on aggregated features or sequential patterns, overlooking complex temporal patterns and interdependencies, whereas traditional process mining flattens the multidimensional nature of real-life behaviors. To bridge these gaps, we explore the application of Object-Centric Process Mining (OCPM) for personal health management. We propose an extension to the standard object-centric event log handling sensor data and demonstrate the feasibility and value of our approach through a case study. This novel OCPM application holds promise for the advancement of personalized digital health interventions and opens new directions for human-centric process mining.

Keywords: Personal Health, Object-Centric Process Mining, Sensor Data

1 Introduction

Amid the increasing global burden of non-communicable diseases, wearables and mobile apps are transforming personal health management by empowering users to actively manage their health and adopt a healthy lifestyle. These technologies monitor health behaviors, such as exercise and sleep, and their dynamic context, promising personalized digital health interventions for behavior change [1]. However, current data-driven methods model a user through aggregated features or sequential patterns, overlooking how human health behaviors evolve and interact in a dynamic context [2].

Recent work has leveraged traditional process mining to capture the process nature of human behaviors, including temporal structures, habits, interleaving, and contextual decisions [3]. Moreover, this approach may enable understanding how users engage in various health behaviors in multiple personal health systems (i.e., wearables and apps). This emerging paradigm is referred to as human-centric process mining. However, traditional process mining requires a single case notion for behavior analysis, flattening the multidimensionality and interdependencies of real-life health processes.

To overcome these limitations, the novel object-centric process mining approach (OCPM) allows modeling a user as a central entity, alongside its interdependent health processes and dynamic context [4]. OCPM was originally conceptualized for business

Pre-print copy of the paper accepted for presentation at the Process Mining Workshops (ICPM 2025 Int. Workshops), to appear in the Springer LNBIP series on Springer Link at https://link.springer.com/

applications, where event logs include activity instances executed in a process. In personal health management, however, events can be activity instances (e.g., self-reports, user-interaction logs) or passive sensor streams (e.g., accelerometer). Recent work in traditional processes mining bridges this abstraction gap from passive sensor events to active behavior events with a pre-processing activity recognition step [3]. In OCPM, we argue that each case notion may require different activity recognition steps and abstraction strategies (e.g., recognize either sleep or physical activity based on accelerometer events). Therefore, raw sensor data should be preserved in object-centric event logs (OCEL) and modelled separately from behavior events to better support activity recognition methods tailored to their time-series nature.

This research introduces the application of OCPM to personal health management by proposing a sensor-augmented OCEL standard (Section 2). We describe the implementation of an OCPM framework through a proof-of-concept, demonstrating feasibility and value of our approach in a case study (Section 3). Lastly, we discuss the implications of our approach for personalized digital health interventions (Section 4).

2 Sensor-Augmented Object Centric Event Log

We propose applying OCPM to personal health management using an augmented four-dimensional OCEL that integrates sensor data and models its relationships between behavior events, objects, and time. Figure 1 shows a meta-model of the sensor-augmented OCEL with key OCPM concepts applied to personal health management. Purple elements highlight our proposed extensions compared to the standard [4]. The proposed meta-model is designed with flexibility in mind, allowing for implementation across various exchange formats. We provide a preliminary JSON schema with validation methods and sample data that adheres to the specific meta-model [5].

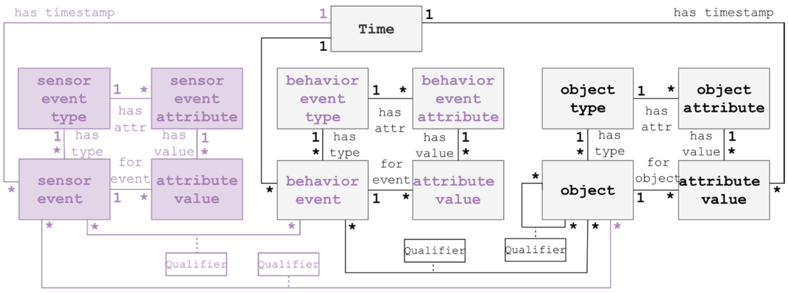

Fig. 1. Sensor-augmented object-centric event data meta-model [4]

Four Dimensions. *Objects* are entities that form case notions characterized by a single *object type.* Objects can represent a user, its health behaviors with lifecycles (e.g., physical activity bouts, sleep episodes), structured time periods (e.g., days, intervention periods) and devices (e.g., a wearable). *Object attributes* may capture their dynamic changes over time. For example, an individual may be described by its dynamic behavior determinants (e.g. self-efficacy, motivation) and contextual factors (e.g., social environment, location, weather). These attributes may explain behavior patterns, such as decision dependencies (e.g., skipping a run due to rain).

Behavior events represent timestamped semantic activity instances across various *types,* including those extracted directly, such as user-system interactions (e.g., content viewing, notifications) and self-reports (e.g., food intake, physical activity log), and those recognized from sensor data (e.g., sleep episode, physical activity bout). Atomic and non-atomic behaviors are abstracted through lifecycle attributes (start/ end). Each behavior event has *attributes* describing the behavior itself (e.g., average heart rate) or its context (e.g., location), which may also be inferred from sensor data.

Sensor events represent timestamped unintelligible measurements captured by sensing hardware in personal health management devices (e.g., smartphone, wearables), typically forming part of a time series. Each event is linked to a single sensor type (e.g. accelerometer) and defined by structured attributes (e.g., x,y,z, for accelerometer).

Time serves as the foundational synchronization dimension across all heterogeneous data, providing single timestamps to events and dynamic object attributes.

Relations. We model the sensor dimension separately to explicitly capture sensor to behavior event (SE2BE) relationships, linking sensor timeseries to multiple behavior events recognized via dedicated timeseries methods. These relationships are many to many and semantically qualified. For example, a timeseries of *accelerometer* events 'recognizes' a *physical activity bout* or a *sleep episode,* while other sensor events stay unlinked. Conversely, behavior events like *self-reports* are independent of sensors. Sensor events, behavior events and objects relate to any number of objects (SE2O, BE2O, O2O) with a semantic qualifier. For example, an *accelerometer* event 'recognizes' a *sleep* event, 'during' a *day* object, and 'belongs to' a *user* object .

3 Proof-of-Concept

We propose and demonstrate the stepwise OCPM framework for personal health management through a proof-of-concept, including tutorial-style code and visuals in a open-source repository [5]. A case study explored the interdependencies between physical activity, perceived stress, and user engagement in self-reporting. A 2-week experiment collected health data from one participant using a device worn on the dominant wrist for at least 12 hours daily. The GameBus-Experiencer app collected: accelerometer data at 50 Hz, photoplethysmography at 12.5 Hz, and stress self-reports using an experience sampling method protocol with up to 5 notifications per day [6].

Step 1: Data Extraction. Data was extracted from the source system to the sensor-augmented OCEL format. This step integrates sensor and behavior data typically stored in fragmented health systems, and prevents redundant data extraction when new case notions emerge [7]. Sensor event types were *accelerometer* and *heart rate*. Behavior event types were *notification* and *stress self-report*. *Notification* objects refer to a *notification* event and the triggered *self-report* event. *Self-report* objects refer to their event. All events and objects were related to a *user* object.

Step 2: Activity Recognition. An activity recognition method transformed sensor data into semantically related behavior events. In this preliminary research, we picked one activity recognition strategy for one case notion, but having sensor data in the sensor-augmented OCEL would enable us to recognize multiple activities with different abstraction strategies for multiple case notions [4]. We recognized *physical activity bouts* using validated threshold decision rules based on accelerometer and heartrate data, as described in the repository. *Physical activity bout* objects with *start* and *end* events were related to *self-report* objects based on temporal proximity, allowing investigating how previous bouts (within 3 hours) affected perceived stress.

Step 3: Data Query. Object-centric queries filtered the sensor-augmented OCEL for a subset of behavior events and objects, forming an OCEL profile. The sensor dimension is omitted, but still represented by the recognized activities. This step enables to drill-down a 3D perspective of interdependent personal health behaviors. We crafted a toy OCEL profile using the PM4PY library to query for specific objects of type *self-report* and their correlated *notification* and *physical activity bout* objects.

Step 4: Process Discovery. The OCEL profile was transformed into an interpretable process model using an OCPM discovery technique. This step enables to model multi-dimensional human health processes revealing complex patterns and end-to-end interactions of a user with a digital intervention. These insights are critical for the personalization of digital interventions. We used the Inductive Miner algorithm and visualized the process as an object-centric Petri Net (OCPN) employing the PM4PY library. Figure 2 illustrates the discovered OCPN capturing the interaction of multiple object types. The process may begin when the user receives a *notification* and may choose to *self-report stress* (decision pattern). The user may receive a *notification* before, during, or after engaging in physical activity (interleaved pattern), affecting perceived stress and response rate. For example, the user may be less likely to *self-report* if a *notification* is received during physical activity.

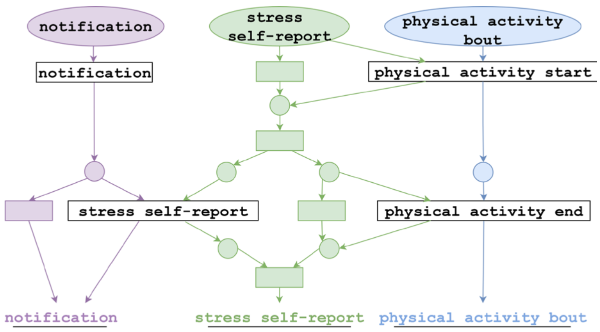

Fig. 2. Discovered Object-Centric Petri Net

4 Discussion

This research introduces OCPM to personal health management, aligning with human-centric paradigms in process mining and healthcare. We augmented the OCEL standard with a sensor dimension and offered a proof-of-concept. Compared to data mining with aggregated features or sequential patterns, process discovery captures complex behavior, including temporal patterns, habits, interleaving, and decision points

[2]. Unlike traditional process mining, OCPM captures the multidimensional and interdependent nature of real-life health behavior and user-system interactions [4].

The key implication of OCPM in personal health management lies in supporting personalized digital interventions. By modeling personal health processes, it can inform just-in-time adaptive interventions that deliver tailored content at opportune moments [1]. Prediction and conformance techniques may guide users from current behavior toward healthier behaviors aligned with improved digital engagement and health goals.

This preliminary work focused on a single user and system, analyzing specific health processes at one abstraction level to demonstrate feasibility. Scaling this approach will require addressing data quality, advancing contextual inference, and adapting discovery techniques for unstructured human behavior. More broadly, the use of personal health data raises important questions about privacy, surveillance, and data governance [7].

As non-communicable diseases continue to increase, personal health management is becoming increasingly vital. Our OCPM framework offers a new direction for modeling personal health behaviors in real-world contexts, laying the groundwork for personalized digital health interventions for behavior change.

Acknowledgments. This publication was AI-edited, with parts of the repository AI-generated. It is part of the project “LIFTS: Healthy Lifestyle for low liTerate teenagerS” with file number KICH1.GZ03.21.011 of the research program “Zorg in eigen leefomgeving”, partly financed by the Dutch Research Council (NWO) under the grant “Kennis- en Innovatieconvenant (KIC), 2020-2023.

References

1. Nahum-Shani, I., et al.: Just-in-Time Adaptive Interventions (JITAIs) in Mobile Health: Key Components and Design Principles for Ongoing Health Behavior Support. Annals of Behavioral Medicine. 52, 446–462 (2018). doi: 10.1007/s12160-016-9830-8.
2. Oyebode, O., Fowles, J., Steeves, D., Orji, R.: Machine Learning Techniques in Adaptive and Personalized Systems for Health and Wellness. International Journal of Human–Computer Interaction. 39, 1938–1962 (2023). doi: 10.1080/10447318.2022.2089085.
3. Brzychczy, E., et al.: Process mining on sensor data: a review of related works. Knowl Inf Syst. 67, 4915–4948 (2025). doi: 10.1007/s10115-024-02297-y.
4. Aalst, V.D., P, W.M.: Object-Centric Process Mining: Unraveling the Fabric of Real Processes. Mathematics. 11, 2691 (2023). doi: 10.3390/math11122691.
5. Ribeiro, M.I., Genga, L., Simons, M., Van Gorp, P.: GameBus-HealthBehaviorMining, https://github.com/MariaInesRibeiro98/GameBus-HealthBehaviorMining.
6. Khanshan, A., Van Gorp, P., Markopoulos, P.: Experiencer: An Open-Source Context-Sensitive Wearable Experience Sampling Tool: 16th EAI International Conference on Pervasive Computing Technologies for Healthcare, PervasiveHealth 2022. Pervasive Computing Technologies for Healthcare. 315–331 (2023). doi: 10.1007/978-3-031-34586-9_21.
7. Auffray, C., et al.: Making sense of big data in health research: Towards an EU action plan. Genome Med. 8, 71 (2016). doi: 10.1186/s13073-016-0323-y.